COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010/Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	7/20/2026	**DATE SUBMITTED:**	7/21/2026
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ACTIVITY NO. 2.1 ARRAYS, POINTERS AND DYNAMIC MEMORY ALLOCATION

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve programming problems using dynamic memory allocation, arrays and pointers

Jenna's Grocery List
Apple PHP 10 x7
Banana PHP 10 x8
Broccoli PHP 60 x12
Lettuce PHP 50 x10

Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.

Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.

```cpp
#include <iostream>
#include <string>

class fruit {
private:
    std::string name;
    double price;
    int qty;

public:
    fruit(std::string n = "", double p = 0, int q = 0) {
        name = n;
        price = p;
        qty = q;
    }

    ~fruit() {}

    fruit(const fruit& f) {
        name = f.name;
        price = f.price;
        qty = f.qty;
    }

    fruit& operator=(const fruit& f) {
        if (this != &f) {
            name = f.name;
            price = f.price;
            qty = f.qty;
        }
        return *this;
    }

    double calculatesum() {
        return price * qty;
    }

    void display() {
        std::cout << "Fruit: " << name
                  << " Price: PHP " << price
                  << " Quantity: " << qty
                  << " Total: PHP " << calculatesum()
                  << std::endl;
    }
};

class veg {
private:
    std::string name;
```

```
    double price;
    int qty;

public:
    veg(std::string n = "", double p = 0, int q = 0) {
        name = n;
        price = p;
        qty = q;
    }

    ~veg() {}

    veg(const veg& v) {
        name = v.name;
        price = v.price;
        qty = v.qty;
    }

    veg& operator=(const veg& v) {
        if (this != &v) {
            name = v.name;
            price = v.price;
            qty = v.qty;
        }
        return *this;
    }

    double calculatesum() {
        return price * qty;
    }

    void display() {
        std::cout << "Vegetable: " << name
                  << " Price: PHP " << price
                  << " Quantity: " << qty
                  << " Total: PHP " << calculatesum()
                  << std::endl;
    }
};

int main() {
    return 0;
}
```

I created the fruit and veg classes with the required constructors, destructor, copy constructor, and copy assignment operator. I also added the attributes and functions needed to store each item's information and calculate its total price.

Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.

```
int main() {

    fruit* app = new fruit("Apple", 10, 7);
    fruit* ban = new fruit("Banana", 10, 8);

    veg* bro = new veg("Broccoli", 60, 12);
    veg* let = new veg("Lettuce", 50, 10);

    fruit* fruits[] = {app, ban};
    veg* vegs[] = {bro, let};

    for (int i = 0; i < 2; i++) {
        fruits[i]->display();
    }

    for (int i = 0; i < 2; i++) {
        vegs[i]->display();
    }

    delete app;
    delete ban;
    delete bro;
    delete let;

    return 0;
}
```

I created dynamically allocated fruit and vegetable objects and stored them in arrays using pointers.

Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in Jenna's Grocery List.

```
double totalsum(fruit* fruits[], int fc,
                veg* vegs[], int vc) {

    double sum = 0;

    for (int i = 0; i < fc; i++) {
        sum += fruits[i]->calculatesum();
    }

    for (int i = 0; i < vc; i++) {
        sum += vegs[i]->calculatesum();
    }

    return sum;
}

int main() {

    fruit* app = new fruit("Apple", 10, 7);
    fruit* ban = new fruit("Banana", 10, 8);

    veg* bro = new veg("Broccoli", 60, 12);
    veg* let = new veg("Lettuce", 50, 10);

    fruit* fruits[] = {app, ban};
    veg* vegs[] = {bro, let};

    std::cout << "Total Amount: PHP "
              << totalsum(fruits, 2, vegs, 2)
              << std::endl;

    delete app;
    delete ban;
    delete bro;
    delete let;

    return 0;
}
```

I created the totalsum() function to calculate the total price of all grocery items. I used loops to add the total cost of each fruit and vegetable, then displayed the final amount.

Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned

```cpp
int main() {

    fruit* app = new fruit("Apple", 10, 7);
    fruit* ban = new fruit("Banana", 10, 8);

    veg* bro = new veg("Broccoli", 60, 12);
    veg* let = new veg("Lettuce", 50, 10);

    delete let;
    let = nullptr;

    std::cout << "Lettuce is removed from the list."
              << std::endl;

    delete app;
    delete ban;
    delete bro;

    return 0;
}
```

I removed the Lettuce object by using the delete operator to free its allocated memory. I also set its pointer to nullptr so it no longer pointed to an invalid memory location.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:
- **Grammar correction**
- **Code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:
- **Helped me fix my grammar into more understandable manner**
- **Assisted me in memory allocation and task explanation**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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